

CANDIDATE LISTS ARE PRIORS: EVIDENCE-FRAME DEPENDENCE IN ACTIVE MODEL SELECTION

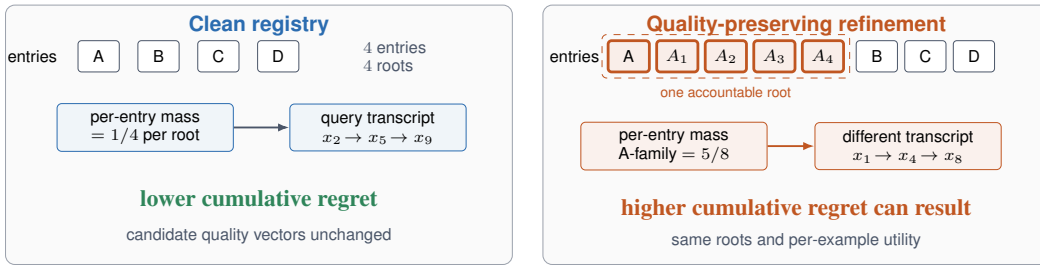
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ABSTRACT

Active model selection uses candidate outputs to decide which costly labels to acquire. In candidate-indexed mechanisms, assigning mass to every listed entry makes the roster itself an unreported experimental prior. We formalize the missing *candidate evidence frame*: accountable roots, an entry-to-root map, and a root-mass rule. A natural exact version alias in a frozen 31-model roster changes all 1,500 paired acquisition paths without adding behavior, although label replacement is small and terminal harm is absent. We prove a tight annotation-budget-linear loss for frame-blind selectors and a constant-size refinement that redirects every query for an exact-match Select-LLM instance. Empirically, response-distinct aliases that preserve correctness coordinate-wise raise cumulative regret by 3.25/3.78/.85 on MedQA/GSM8K/OpenBookQA and terminal regret by 2.91/10.56 points on the first two. Fixed-query effects are zero, isolating acquisition mediation; order and tie audits preserve the result. Raw-text adapters provide a separate holdout-reference-free bridge: a development-informed, disjoint 13,000-row IMDB outcome yields +.702 [.615, .788] terminal points under the same acquisition/evidence view. A one-shot Yelp transfer is directionally positive but misses its predeclared materiality threshold. Other reference-free and cross-selector results are selective, and untargeted additions are mostly benign. We therefore establish targeted evidence-frame dependence, not universal instability, live-registry admission, or production compromise.

1 INTRODUCTION



Predictions expose submitted response rows, but not the accountable map $\rho : E \rightarrow R$ or which entries deserve independent evidential mass.

Figure 1: The candidate list is an empirical prior. Adding entries from one accountable root can change active queries even when the added entries introduce no new per-example quality vector. The observable responses do not authenticate the root map. This schematic illustrates the mechanism; Tables 5 and 9 report measured effects.

Expert labels are costly, so DiffUse, MODEL SELECTOR, CODA, LLM Selector, and Select-LLM use candidate outputs, weak judges, or consensus to prioritize labels for model comparison (Ashury Tahan et al., 2024; Okanovic et al., 2025; Kay et al., 2025; Durmazkeser et al., 2025; 2026). In the candidate-indexed Bayesian and consensus mechanisms we audit, however, every listed entry

receives separate probability or consensus mass. They state no invariance requirement for two lists encoding the same behavioral alternatives.

The missing object is the *candidate evidence frame*: a map from listed *submitted entries* to accountable *candidate roots*, together with a root-mass rule. Predictions alone need not reveal the ownership or lineage evidence needed to authenticate this frame. Consequently, treating entries as the population makes registry representation part of the experiment.

The issue occurs naturally. In our frozen roster, `claude-instant-1.2` and `claude-instant-v1` have identical responses and scores on all 2,500 task items. Restoring the latter to its 30-entry quotient raises one behavior’s mass from $1/30$ to $2/31$ and changes every one of 1,500 Select-LLM paths; random querying is unchanged. Because unique-label substitution is only 8.7/4.7/3.2% of budget and terminal harm is absent, this is evidence of representation dependence, not broad natural damage.

We organize the paper around three questions. *Is the frame necessary?* We prove a tight budget-linear cost for policies that cannot observe it and construct four aliases that redirect every query in a Select-LLM instance. *Does refinement causally change model-selection cost?* Public-reference T2 aliases preserve correctness coordinate-wise while all existing candidates, labels, pools, and code remain fixed. They increase cumulative regret on three tasks and terminal regret by 2.91/10.56 points on MedQA/GSM8K; a fixed-query counterfactual removes the effect. *Can the channel operate without target outcomes?* Calibration-only variants pass three of eight untouched scenarios, and raw-text learned adapters yield $+ .702 [+ .615, + .788]$ terminal points on a disjoint 13,000-row IMDB outcome. A stricter one-shot Yelp transfer is positive but below its predeclared materiality threshold. Cross-selector and untargeted audits show selective rather than generic degradation.

Sequential multiplicity is known in strategic-replication bandits, perturbed-clone active evaluation, redundant-committee quotients, and static leaderboards (Shin et al., 2022; 2025; Lanctot et al., 2026; Nguyen et al., 2025; Singh et al., 2025; Hays et al., 2026). We claim neither the first sequential clone effect nor the first multiplicity attack. Our remaining object is specific: in shared-label model selection with a latent root map, entry mass changes which reusable references are acquired and hence root-level regret. A static rank can be recomputed after quotienting; an acquisition transcript cannot recover labels never requested. Exact hashes remove literal copies, while response similarity cannot authenticate ownership.

Our contributions are:

1. We define evidence-frame ambiguity and show both its information-theoretic cost and a constant-size, full-path amplification.
2. Exact-quality interventions and fixed-query counterfactuals establish the complete refinement $\rightarrow$ acquisition $\rightarrow$ evidence $\rightarrow$ root-cost chain; a natural alias anchors the representation problem in an existing roster.
3. Reference-free, learned, cross-selector, defense, and untargeted audits identify where the effect transfers and where it does not. They support a targeted evaluation-integrity claim, not universal or production harm.

We do not claim that every active selector is vulnerable, that a natural alias is adversarial, that every refinement is harmful, or that similarity reveals ownership. The resulting reliability requirement is narrower: an active evaluation should either bind submitted entries to a declared candidate evidence frame or qualify conclusions that depend on unaudited entry multiplicity.

2 SETTING AND CANDIDATE EVIDENCE FRAMES

Active selection. Let $\mathcal{X} = \{x_1, \dots, x_N\}$ be an unlabeled evaluation pool and $E = \{e_1, \dots, e_H\}$ the submitted candidate entries. Entry e exposes response vector $a_e = (a_e(x_1), \dots, a_e(x_N))$ and has oracle utility vector u_e , binary per-example correctness in our experiments. An active policy π sequentially acquires B references. After step t , its transcript contains the queried indices, acquired references, and selected entries; using an oracle root map, we evaluate the corresponding root-level choices $\hat{r}_1, \dots, \hat{r}_t$.

Definition 1 (Candidate evidence frame). A frame $\Phi = (R, \rho, w)$ maps submitted entries to accountable candidate roots through $\rho : E \rightarrow R$ and assigns root mass w_r . A root is the unit that the evaluation claim intends to count once; it need not be observable or cryptographically authenticated. Two frames are response-compatible when they expose the same entry identifiers and response vectors.

Definition 2 (Registry refinement). A refinement of root r adds entries mapped to r without modifying existing entries. It is exact when an added response vector copies its parent and quality-equivalent when its oracle utility vector equals the parent’s coordinate-wise, even if their responses differ. Root-level refinement integrity requires a declared within-root rule under which such representation changes do not silently redefine the evaluation claim.

For example, a uniform per-entry mechanism induces the empirical measure $\mu_E = |E|^{-1} \sum_{e \in E} \delta_{a_e}$. Adding k entries changes one root’s represented mass without adding a root; candidate-consensus mechanisms can change analogously because consensus is computed over listed entries. A policy is *frame-blind* when it observes responses and acquired references but not authenticated ρ or w .

Table 1 compares strategic-replication bandits (Shin et al., 2022; 2025), active evaluation and selection (Lanctot et al., 2026; Liang et al., 2020; Nguyen et al., 2025; Okanovic et al., 2025; Kay et al., 2025; Durmazkeser et al., 2025; 2026), arena candidacy (Singh et al., 2025; Huang et al., 2026; Hays et al., 2026), clone weighting (George, 2010; Procaccia et al., 2025; Berriaud & Wattenhofer, 2026a), false-name work (Douceur, 2002; Burnat & Davidson, 2026), and lineage (Shang et al., 2026).

Table 1: Closest-work obligation boundary. These lines solve important pieces, but not the effect of an unaudited candidate unit on a complete adaptive selection transcript.

Line	Unit or evidence	Path?	Unmet obligation here
Replication bandits	Registered arms with known agent ownership	Yes	Shared labels, latent roots, terminal model choice
Sequential selection	Perturbed clones, pattern quotients, exact mergers, or supplied IDs	Yes	Shared labels, latent roots, exact-utility refinement, mediation
Arena candidacy	Provider/model clones	Final rank	Task-label acquisition transcript
Dilution/clone weights	Fixed neighborhood or metric	No	Root evidence and adaptive transcript
False-name/Sybil	Authority, resource, or evidence cluster	Not here	Active candidate-selection consequence
Model lineage	Ancestry evidence	No	Candidate mass and within-root rule

We do not claim the first exact-duplicate preprocessor, sequential replication or clone effect, registry effect, multiplicity attack, clone-proof weight, evidence-backed quotient, or lineage system. Our object is the shared-label sequential boundary: what predictions justify about response-distinct candidate units, how latent same-root multiplicity alters acquired reusable labels, and what root-level decision cost follows. Lineage is useful missing input, not a selection policy supplied here.

Natural refinement versus attack. Our natural anchor compares the original frozen roster with the exact-response quotient that removes its later-index version alias. It attributes neither common control outside the evaluation frame nor adversarial intent to the provider. It asks only whether a behaviorally redundant listed entry changes the active experiment. The adversarial tiers below are separate interventions that test whether an actor controlling one root can exploit the same entry-mass channel.

Offline threat tiers. The attacker controls one root and registers before acquisition. **T0** adds exact copies as a mechanism and defense control. **T1** learns on disjoint calibration references, then uses only holdout prompts, options, and parent outputs; quality is measured and approximately bounded, not assumed equivalent. **T2**, the main causal intervention, uses public or independently acquired references and changes only parent-wrong responses, exactly preserving correctness. Thus T2 fixes quality but assumes such reference access; T1 removes holdout-reference access and tests transfer. Across tiers, the attacker cannot modify existing entries, references, selector code, pool, budget, temperature, seeds, or tie-breaking, or adapt after acquisition begins.

Let $\ell(r)$ be root r 's full-pool loss and r^* the best root. We report ordered-path divergence, terminal regret $\ell(\hat{r}_B) - \ell(r^*)$, and the anytime quantity

$$\text{CReg}_B = \sum_{t=1}^B [\ell(\hat{r}_t) - \ell(r^*)].$$

Cumulative regret models repeated interim use; pairing it with terminal regret prevents a transient path effect from being misreported as a worse final choice.

3 FRAME IDENTIFIABILITY AND SEQUENTIAL AMPLIFICATION

We ask whether labels can compensate for a missing frame in any response-only policy, and whether a constant refinement can redirect a current selector. The first result establishes information necessity; the second gives concrete amplification.

Response-compatible worlds. Let $O(E) = \{(e, a_e) : e \in E\}$ be the public response view. Frames exposing the same $O(E)$ and label transcript are indistinguishable to a frame-blind policy: labels reveal useful behavior, not whether entries are independent roots. A fresh hidden-frame choice at every step makes this ambiguity accumulate.

For the construction below, let $V_{\Phi, B}^\pi$ be the expected information, in bits, acquired by policy π about B latent binary candidate coordinates, and let $V_{\Phi, B}^*$ be the optimum for a policy that knows Φ . Define $I_{\text{hi}} = 0.540852$ and $I_{\text{lo}} = 0.459148$ bits.

Theorem 1 (Tight cost of a missing evidence frame). *For every integer budget $B \geq 1$, there is one fixed public response registry, a pool with two binary queries per coordinate, and 2^B response-compatible frames such that every possibly randomized frame-blind policy has, on some fixed compatible frame,*

$$V_{\Phi, B}^* - V_{\Phi, B}^\pi \geq \frac{B}{2} (I_{\text{hi}} - I_{\text{lo}}) = 0.040852B \text{ bits}.$$

The bound is tight. The corresponding tight minimax excess coordinate-wise Hamming risk is $B/24$.

Proof intuition. One latent frame bit per coordinate selects which of two query polarities yields I_{hi} instead of I_{lo} . Under uniform compatible frames, an untouched bit is independent of prior observations, so a frame-blind policy chooses the informative polarity with probability at most one half. Revisiting cannot improve its minimax value. Averaging yields a fixed hard frame; independent randomization attains equality. Appendix B gives the construction and $1/24$ Hamming-risk calculation.

The construction grows exponentially with B , proving necessity rather than practical registry size. A constant change can nevertheless be amplified by a concrete selector.

Theorem 2 (Constant-registry full-path amplification). *For every odd $B \geq 3$ and finite Select-LLM temperature $\tau > 0$, there is a pool with five canonical response behaviors and $2B$ items such that adding four exact aliases produces nine submitted entries, redirects all B acquisitions to a disjoint query block, and changes cumulative regret from zero to at least*

$$\frac{B^2 - 1}{4B} = \Omega(B),$$

while the terminal regret is $(B - 1)/(2B)$. The two best canonical behaviors and all existing responses are unchanged.

Proof intuition. The clean registry always prefers a good query block. Four aliases move another behavior's mass above one half along an alternating-label path, making every bad query preferable and reinforcing the condition after each selection. Counting odd steps gives both regret bounds; Appendix B verifies them for every finite τ .

This is mechanism-specific, not universal harm, and exact hashing blocks the literal construction. Section 4 therefore tests response-distinct, quality-equivalent variants.

Proposition 1 (Selector-agnostic root factorization). *Suppose ρ is authenticated, each root has fixed mass w_r , and at every step a selector’s state, action kernel, and decision depend on entries only through root summaries S_r . If S_r is invariant to an allowed within-root refinement, then the distribution of the complete query transcript and root decisions is invariant to that refinement.*

Transcript induction proves the claim (Appendix B). It covers uncertainty, diversity, and mutual-information rules only when they actually factor through authenticated, refinement-invariant root summaries; a method label alone gives no protection. Exact quotienting handles unique response patterns, not response-distinct variants. Predictions also cannot authenticate ρ , so deployment still requires a declared canonical entry or fixed within-root mixture.

4 CAUSAL INTERVENTION AND EVALUATION

Primary mechanism. Select-LLM uses one similarity s for both reference–output posterior evidence and output–output acquisition (Durmazkeser et al., 2026). Our independent exact-answer specialization sets $s(a, b) = \mathbf{1}[a = b]$ in both places:

$$p_t(e) \propto \exp\left(\tau^{-1} \sum_{s < t} s(r_{i_s}, a_e(x_{i_s}))\right), \quad A_t(x) = \sum_y \left(\sum_{e: a_e(x)=y} p_t(e) \right)^2.$$

The selector queries the unlabelled item minimizing A_t , observes its reference, and chooses the root with maximum accumulated exact-match evidence. This source-fidelity statement applies to the natural anchor and primary T2 experiment; transfer experiments that deliberately separate acquisition and evidence views are labeled as such. V2 canonicalizes entries by full-response SHA-256, root ID, and label hash, so submitted-list order cannot determine a paired result.

We use frozen HELM-Lite v1.0.0 responses and reference scores for 31 models (Liang et al., 2023) on MedQA, GSM8K, and OpenBookQA (Jin et al., 2021; Cobbe et al., 2021; Mihaylov et al., 2018). Table 2 gives the primary conditions.

Table 2: Frozen primary protocol. Each reported condition uses 500 paired pool draws.

Task	N	Pool	Budgets	τ	Development-locked parent
MedQA	1000	500	10/30/50	3	GPT-4-1106-preview
GSM8K	1000	500	10/20/30	5	GPT-4-0613
OpenBookQA	500	400	10/20/30	5	text-davinci-002

Evidence ladder. The experiments answer different questions and are not pooled as interchangeable attacks. Table 3 states the role and information boundary of each layer.

Table 3: Evidence layers. “Outcome-free” means that the deployed alias receives no evaluation reference or item lookup.

Layer	Refinement	Outcome access	Scientific role
Natural anchor	Existing exact version alias	None	Occurrence and representation non-invariance
T2 causal test	Response-distinct, exact utility	Public/acquired references at construction	Isolate candidate-mass causality
T1 transfer	Calibration-only abstention variants	No holdout references	Prospective cross-task bridge
Learned endpoint	Raw-text error adapters	Train/development only; outcome sealed	Outcome-free executable same- s bridge
Cross-selector	Official released score view	Strong-judge row unchanged	Test a second acquisition family

Natural and quality-equivalent refinements. Before selector outcomes, we identify the unique pair whose complete response and utility rows match on all three tasks. The 30-entry quotient removes the later version; the natural roster restores it and maps both entries to one root. Pair discovery uses row equality, not selector outcomes, and a locked replay distinguishes changed order from changed query sets (Appendix D).

The main T2-A constructor adds four response-distinct aliases. A frozen hash chooses parent-wrong coordinates; each alias substitutes another wrong multiple-choice option or a nonzero numeric offset from the reference. Hence $u_{e'} = u_e$ coordinate-wise while response hashes differ. Construction reads the parent output, public item information, and references, but no peer response, pool, or trajectory. Targets are selected on disjoint development pools and replayed without retargeting on 500 fresh pools per task. Prompt-only runtime wrappers receive no item ID, reference, peer output, pool seed, or selector state (Appendix C). T2 therefore identifies a causal channel under its stated reference-access tier; it is not evidence of hidden-label admission.

Outcome-free transfer. T1 freezes eight untouched HELM-Lite QA/translation scenarios before their holdout outcomes. Calibration-only TF-IDF error scores define four abstention wrappers that receive no holdout reference, utility, peer output, pool, or selector state. Because these tasks use exact-response grouping for acquisition but F1/BLEU for evidence, T1 is explicitly a decoupled-view audit rather than source-faithful Select-LLM (Appendix A).

The learned audits remove that view mismatch: a raw-input adapter emits its parent’s hard label or an out-of-label abstention token, and the same literal exact-match function supplies acquisition and reference evidence. Earlier terminal-null and stopped studies remain in Appendices I–L. The main learned result uses a hash split of IMDB into 12,000 development and 13,000 one-time outcome rows. A complete development grid and a disjoint robustness rule select one condition; all label-free held-out predictions, model weights, thresholds, code, seeds, and outcome hash are bound before the outcome opens. This is development-informed held-out confirmation, not before-all-data preregistration (Appendix M).

A stricter Yelp audit freezes a public-download-based four-checkpoint roster, the inherited adapter/threshold recipe, one selector condition, and a no-replacement rule before loading the target data. It contains no target selector search: fixed train/calibration/safety partitions fit the adapters, and all predictions on the complete 38,000-row test are bound before the separately sealed outcome opens once. IMDB and Yelp test different registries within binary sentiment; Yelp is not broad task-family transfer (Appendix N).

Causal controls and inference. For every seed, clean and refined runs share the pool, references, budget, temperature, and tie rule. We report paired cumulative and terminal root regret, ordered-path and query-set change, and final-root transitions. Three controls isolate interpretation: candidate-invariant random queries test whether active acquisition is nonvacuous; a fixed query matrix separates acquisition-mediated from static registry effects; and 16 registry permutations plus all nine exact query/root tie policies test implementation artifacts. Paired bootstrap intervals use 10,000 repetitions, sign-flip tests are paired, and prespecified families receive Benjamini–Hochberg correction. Complete temperature, budget, alias-count, and distance grids retain all eligible cells and all negative outcomes (Appendix C).

Defenses and second selector. We compare exact quotienting, distance-matched behavioral covers, authenticated-root mass, and three clone-robust graph weights (Berriaud & Wattenhofer, 2026b). Each defense is evaluated jointly on residual attack effect and clean cumulative/terminal cost; prediction-only similarity is never treated as proof of ownership (Appendix P). We also retain an official CODA audit and a preregistered audit of all six official LLM Selector datasets. The latter preserves every strong-judge row, modifies only one parent’s weak-judge views, and requires exact fixed-query equality and corrected terminal harm. It is a released score-matrix intervention, not raw-endpoint admission (Appendices E and F).

5 RESULTS

Table 4 separates occurrence, causal identification, and transfer. These layers answer different questions: a natural alias need not cause harm, and a controlled causal effect need not constitute a production attack.

Table 4: Result ladder. Terminal “passes” use each audit’s frozen gate; all reported causal terminal effects have zero fixed-query effect.

Evidence	Scope	Terminal result	What it establishes
Natural exact alias	3 tasks, 1,500 runs	No reliable harm	Representation non-invariance
Exact-quality T2-A	3 tasks	Harm on 2/3; CReg on 3/3	Acquisition-mediated causality
Calibration-only T1	8 untouched scenarios	Gate passes on 3/8	Selective reference-free transfer
Learned IMDB	13,000 held-out rows	+ .702 [.615, .788] pp	Executable same-view terminal effect
Prospective Yelp	38,000 one-shot rows	+ .353 [.309, .398] pp; below gate	Directional no-search transfer
Official LLM Selector	6 datasets	Gate passes on 2/6	Second-family transfer

5.1 REPRESENTATION DEPENDENCE OCCURS NATURALLY WITHOUT RELIABLE NATURAL HARM

The two frozen version entries are the unique cross-task exact response-and-utility pair: zero mismatches over 2,500 items and equal task qualities .548/.721/.844. Restoring the later entry to its 30-entry quotient adds no behavior or quality vector, yet changes at least one query in all 500 paired runs per task. Mean changed positions and query-set Jaccards are 40.744/.844, 21.752/.911, and 16.010/.940 (Table 5, Panel A). Candidate-invariant random queries produce identical query matrices, root histories, and regret, so the active effect follows from changing one behavior’s entry mass from 1/30 to 2/31.

Table 5: Natural and targeted refinements on 500 paired pools per task. Panel A restores one naturally occurring exact version alias to its 30-entry quotient. Panel B reports the canonical V2 replay of fresh-pool, 5%-distance T2-A effects constructed without peer responses. Every run changes path; CReg is summed through B and positive is worse.

<i>A. Natural exact alias: occurrence and representation non-invariance</i>					
Task	Changed positions	Query Jaccard	Δ CReg [95% CI]	Unique repl.	
MedQA	40.744/50	.8437	−0.5002 [−0.6152, −0.3883]	4.358	(8.72%)
GSM8K	21.752/30	.9111	+0.0593 [0.0275, 0.0921]	1.420	(4.73%)
OpenBookQA	16.010/30	.9395	−0.0276 [−0.0634, 0.0083]	0.960	(3.20%)
<i>B. Target-locked, peer-response-free T2-A: targeted exploitability</i>					
Task	B	Δ CReg [95% CI]	Δ terminal pp [95% CI]	Harmed pools	
MedQA	50	+3.2518 [3.0214, 3.4842]	+2.906 [2.316, 3.543]	92.0%	
GSM8K	30	+3.7811 [3.6177, 3.9433]	+10.559 [9.908, 11.234]	99.2%	
OpenBookQA	30	+0.8459 [0.7650, 0.9260]	+0.006 [−0.016, 0.031]	78.8%	

Ordered displacement is not the same as acquiring many new labels. Query sets change in 94.0/87.4/66.0% of runs, but only 4.358/1.420/.960 unique labels are replaced (8.72/4.73/3.20% of budget). Cumulative-regret directions mix and every terminal interval contains zero. The natural anchor therefore establishes occurrence and limited evidence substitution, not adversarial intent or natural damage.

5.2 QUALITY-PRESERVING REFINEMENT COMPLETES THE CAUSAL CHAIN

Every T2-A alias has a distinct response hash but exactly the parent’s correctness vector. Existing entries, references, best root, pool distribution, and selector remain fixed, and the constructor reads no peer response. All 500 fresh trajectories change on every task. Cumulative regret rises by 3.252/3.781/.846 (Table 5, Panel B; corrected $q = 9.999 \times 10^{-5}$); terminal regret rises by 2.91/10.56 points on MedQA/GSM8K, while OpenBookQA is cumulative-only.

The counterfactual identifies the mediator: forcing clean and refined registries to use the same query matrix makes root histories and regret exactly equal. Thus adding lower-quality roots or changing a static deployment rule cannot explain the effect. All 48,000 replays across 16 registry permutations are bitwise canonical-equal, and every cell in the nine-policy query/root tie cross retains positive cumulative-regret intervals. Clean active selection also beats random querying on two tasks, excluding a vacuous selector.

The effect is selectable rather than monotone. Development-to-final target-effect correlations are .991/.995/.999, but complete target medians are $-.006/-.046/-.176$. All primary temperature and budget cells are positive; the wider OpenBookQA grid contains 171 positive and 69 negative cells. Candidate refinement can steer the trajectory in either direction, and the paper’s claim concerns the existence of targeted harm.

5.3 OUTCOME-FREE EXECUTION IS REAL BUT TRANSFERS SELECTIVELY

Calibration-only T1 aliases pass terminal and active-minus-fixed gates on NarrativeQA and WMT14 fr-en/ru-en (3/8; corrected $q \leq .00027$). Three nonpasses improve and two are null. This is prospective cross-task evidence, not a universal success rate.

The stronger learned confirmation uses one literal exact-match view for both acquisition and evidence. Four independently trained IMDB raw-text adapters emit the parent label or abstain, never improve a coordinate, and lose only .438–.523 accuracy points. On the disjoint 13,000-row outcome, 99.93% of paths and 60.67% of final roots change. Terminal and active-minus-fixed regret rise by $+.702$ [.615, .788] points, cumulative regret by $+.02795$ [.02467, .03129], and fixed-query effects are zero. Because the task, roster, and condition were developed on separate IMDB rows, this is held-out outcome confirmation rather than untouched-task transfer or live admission.

Yelp tests that stricter boundary. A public-checkpoint roster and inherited pipeline are fixed before target access, with no target selector grid or replacement outcome. Aliases lose .376–.561 accuracy points; 99.90% of paths and 41.70% of final roots change. Terminal and active-minus-fixed regret rise by $+.353$ [.309, .398] points and fixed-query effects remain zero, but the effect misses the predeclared .5-point materiality threshold. We therefore retain it as a negative, directional one-shot transfer rather than a second primary success.

5.4 DEFENSES AND SECOND-FAMILY RESULTS BOUND THE CLAIM

Exact quotienting misses response-distinct aliases. A distance-matched 5% cover suppresses T2-M, but larger radii merge natural models and can cost 3.975 terminal points. Soft clone-robust weights mitigate 74/105 cells, yet no fixed response-only setting is non-worsening across all attacks and clean conditions. Authenticated lineage is exact because it supplies the missing root information (Appendix P).

The official LLM Selector audit passes terminal gates on AlpacaEval and Bingo (2/6; .321/.498 points, corrected $q = 3.0 \times 10^{-5}$), with zero fixed-query effects. The remaining datasets improve, miss correction, or are null; CODA is mixed. Only 8/420 untargeted additions are materially harmful versus 27 beneficial, and a preregistered fragility score fails five of six gates. Together, these results support a cross-family but targeted failure, not average harm (Appendices F, E, and O).

6 RELATED WORK

Closest conceptually, strategic-replication bandits let known agents register duplicate arms and motivate agent-first defenses (Shin et al., 2022; 2025). They observe arm-local rewards under supplied ownership; here roots are latent, one label scores every entry, and selection ends at a root. Active evaluation tests separately scored perturbed clones; UNREAL and PUTSC quotient redundant committees or candidates (Lanctot et al., 2026; Nguyen et al., 2025; Liang et al., 2020). We claim neither the first sequential replication nor clone effect. Our remaining object is shared-label exact-utility refinement with fixed-query mediation to root regret (Table 1); correlation-aware selection fixes an ensemble (Turkmen et al., 2026).

Static registry attacks alter ranks, models, matches, or descriptions, not future labels (Hays et al., 2026; Suri et al., 2025; Singh et al., 2025; Huang et al., 2026). Dilution priors, clone weights, quotients, and lineage constrain static influence or identify roots (George, 2010; Berriaud & Wattenhofer, 2026b; Procaccia et al., 2025; Burnat & Davidson, 2026; Shang et al., 2026); ours changes which reusable labels are acquired, which post-hoc quotienting cannot recover.

7 LIMITATIONS, IMPLICATIONS, AND CONCLUSION

What is established. Candidate-indexed mass can make a shared-label active experiment depend on roster representation. Exact-quality T2-A and fixed-query equality establish the complete refinement-to-root-cost causal chain, and learned raw-input adapters show that holdout-reference lookup is not necessary for a terminal effect. The theory explains why responses alone cannot generally recover the missing frame.

What is not established. The natural anchor is one pair and causes no reliable terminal harm. T2-A uses public or independently acquired references; T1 passes 3/8, LLM Selector 2/6, and untargeted additions are mostly benign. Most learned same-*s* tasks are terminal-null or stop. IMDB is disjoint held-out evidence, but its condition was developed on 12,000 rows of the same benchmark; prospective Yelp misses its .5-point materiality gate, and both are binary sentiment. The adapters learn separate weights while sharing public base architectures, and no endpoint is admitted to a live registry. We therefore do not establish production compromise, clinical harm, universal terminal damage, adversarial intent, independent full-base checkpoints, or the first sequential clone effect.

Implication. The operational requirement is not a universal deduplication metric. An evaluator must declare what a root means for its claim, bind entries to that unit using appropriate provenance or governance evidence, assign fixed root mass, and specify a refinement-invariant within-root summary (Proposition 1). When authentication is unavailable, multiplicity sensitivity is part of the uncertainty of the reported selection result and should be audited explicitly.

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AI USE STATEMENT

Generative AI tools assisted topic and gap discovery, conceptual framing and hypothesis refinement, literature discovery and organization, methodology and experiment-design critique, code drafting and experiment orchestration, formulation and checking of formal statements and proof drafts, result analysis and interpretation, adversarial manuscript review, artifact documentation, and language editing. AI outputs were not treated as sources of evidence or as autonomous authorship. The authors manually checked citations against primary sources, executed and validated the code and locked outputs, inspected the proof obligations, reviewed all AI-assisted text, and take responsibility for the complete submission.

ETHICS STATEMENT

The work audits public benchmark responses and does not use clinical records or interact with deployed patients. We disclose attack prerequisites and negative results to avoid overstating private-label or clinical risk. The intended use is to improve the integrity of evaluation registries and active selection protocols.

REPRODUCIBILITY STATEMENT

The anonymous artifact records the natural-alias lock, preregistered targets, paired seeds, complete robustness and defense grids, official-selector provenance, and machine-readable outputs. Its manifest binds the learned-audit history through Step 105, including the IMDB held-out primary and the prospective Yelp NO_GO. It includes learned weights, pre-outcome prediction locks, sealed-outcome bindings, raw 3,000-pair trajectories, literal status labels, and independent reconstructions. Offline mode verifies every bundled byte and anonymity. Full mode rebuilds the paper and reconstructs both final studies’ trajectories, inference, and gates; optional raw-input mode re-infers their roots and aliases when pinned public base weights are available. Public source packages, base weights, and third-party repositories are not embedded, and historical validators are not all rerun transitively.

A PROSPECTIVE REFERENCE-FREE TERMINAL AUDIT

Outcome-blind task freeze. An initial preregistration named 18 fresh MMLU subjects, but all 31 model-prefix requests for its first frozen subject returned HTTP 404 before any response or outcome was observed; HELM-Lite v1.0.0 contains only the five MMLU subjects already used elsewhere in the project. We retained that structural failure rather than replacing a result. A metadata-only enumeration then identified eight previously untouched scenarios with the complete 31-model roster. Before downloading their content or outcomes, a separate protocol froze NarrativeQA, NaturalQA closed/open, and WMT14 cs/de/fr/hi/ru-en. These exhaust the untouched, structurally complete non-LegalBench HELM-Lite scenarios left after excluding prior tasks; no frozen scenario could be dropped or replaced.

Each task was split by a manifest-keyed SHA-256 ordering: NarrativeQA contributes 235 calibration and 235 sealed-holdout items, and every other scenario contributes 500 and 500. The first metric present for every model-item pair under the frozen order F1, quasi-exact match, exact match, BLEU-4, BLEU was selected without attack outcomes; this yields F1 for the three QA scenarios and BLEU-4 for translation. Utilities lie in $[0, 1]$.

Attack freeze and gate. Five-fold out-of-fold TF-IDF bigram ridge prediction estimates the selected parent’s continuous error from prompt and parent output. A prediction-only exact-clone screen retains five parents. Calibration then selects among three support-overlap rules, distances .5/1%, budgets 10/20/40, and clean-selected temperatures 1/3/5 by maximizing active-minus-fixed terminal harm over 200 paired pools. Only after that choice does the builder open the selected parent’s blind holdout prompts and outputs. It changes the top predicted-error coordinates to four alias-specific opaque abstention strings; no holdout reference, utility, peer output, pool, trajectory, or selector state is read. Unchanged utilities are inherited and changed utilities are zero, so every alias is coordinate-wise non-improving and loses at most its declared distance. The tokens are subsequently checked against all sealed references and have zero collision.

Stage B runs exactly 1,000 paired pools (seeds 88400–89399) per frozen scenario. A scenario passes only if (i) all aliases remain within one percentage point, (ii) both active terminal harm and active-minus-fixed terminal harm have paired-bootstrap 95% lower bounds above zero, and (iii) their separately eight-task BH-adjusted one-sided sign-flip q -values are below .05. Fixed-query terminal effects are reported separately; any nonzero value forbids exact mediation. At least two passing scenarios were required for promotion.

Table 6: Prospective reference-free terminal audit on eight previously untouched HELM-Lite scenarios. Values are utility percentage points; q^* is the larger of the two eight-task BH-adjusted one-sided sign-flip q -values for active terminal harm and active-minus-fixed terminal harm. Clean and refined fixed-query effects are exactly zero in every row. All rows were frozen and retained.

Scenario (metric)	d	B	Quality gap	Δ terminal [95% CI]	q^*	Path/root change	Gate
NarrativeQA (F1)	1%	10	-.851	+3.021 [2.766, 3.282]	.00027	100.0/45.6%	Pass
NaturalQA closed (F1)	.5%	10	.000	-1.213 [-1.443, -.985]	1.000	97.6/55.5%	No
NaturalQA open (F1)	1%	10	-.842	-.317 [-.552, -.075]	1.000	99.6/35.6%	No
WMT14 cs-en (BLEU-4)	1%	10	-.109	+.080 [-.033, .192]	.161	99.1/39.5%	No
WMT14 de-en (BLEU-4)	1%	10	-.275	-.087 [-.169, -.002]	1.000	94.2/35.9%	No
WMT14 fr-en (BLEU-4)	1%	40	-.064	+.139 [+ .102, .177]	.00027	75.1/10.6%	Pass
WMT14 hi-en (BLEU-4)	1%	20	.000	+.001 [-.033, .035]	.757	100.0/15.4%	No
WMT14 ru-en (BLEU-4)	1%	10	.000	+.250 [+ .154, .350]	.00027	97.4/46.1%	Pass

Three scenarios pass: NarrativeQA, WMT14 fr-en, and WMT14 ru-en. Their worst alias quality gaps are -.851, -.064, and .000 percentage points, respectively; thus the Russian pair is exact-utility-equivalent on the sealed holdout. All eight fixed-query root histories and regret arrays are byte-identical, making the active and active-minus-fixed point effects equal. The five nonpassing scenarios are not hidden: two confidence intervals cross zero, while NaturalQA closed/open and WMT14 de-en significantly improve the selected terminal utility. The machine decision is PROMOTE_REFERENCE_FREE_TERMINAL_BRIDGE; it supports a selectable, reference-free attack on new task families, not monotone transfer.

The Stage-A attack was frozen before deserializing the sealed outcomes. The Stage-B lock binds the preregistration, attack rows, sealed package, runner, independent validator, and six runtime dependencies. A separate implementation reconstructs canonicalization, active and fixed acquisition loops, every raw path/regret array, 10,000-repetition intervals and sign-flip tests, BH adjustment, quality/integrity gates, and the final 3/8 decision. All eight task checks, all four lock checks, and the decision check pass under `PASS_STEP87B_INDEPENDENT_RECONSTRUCTION`. The immutable results and raw-array SHA-256 values begin `FC953C87` and `26422130`; the independent-validation hash begins `1EF6C0F7`.

B FULL PROOFS AND FORMAL VERIFICATION

B.1 PRODUCT-FRAME CONSTRUCTION

Fix budget B and let each canonical behavior be a vector $z \in \{A, B\}^B$. The public registry contains 2^B exact submitted copies of every z , for 4^B submitted IDs in total. A latent frame is indexed by $\omega \in \{A, B\}^B$. Define

$$m_\omega(z) = |\{t : z_t = \omega_t\}|.$$

Frame Φ_ω partitions the 2^B public copies of z into $r_\omega(z) = 2^{m_\omega(z)}$ accountable roots, each containing $2^{B-m_\omega(z)}$ aliases. These are integer partitions of the same public leaves. Every frame contains

$$\sum_z 2^{m_\omega(z)} = 3^B$$

roots. A uniform root prior therefore induces

$$\Pr_\omega(F = z) = \frac{2^{m_\omega(z)}}{3^B} = \prod_{t=1}^B \begin{cases} 2/3, & z_t = \omega_t, \\ 1/3, & z_t \neq \omega_t. \end{cases}$$

The candidate coordinates are independent, with $\Pr(F_t = \omega_t) = 2/3$, although the submitted IDs and responses are identical in all 2^B frames.

For each coordinate t , the fixed pool contains two binary queries $q_{t,+}$ and $q_{t,-}$. With $c = 1/4$, their channels are

$$\begin{aligned} q_{t,+} : \quad & \Pr(Y = 1 \mid F_t = A) = 1, & \Pr(Y = 1 \mid F_t = B) = c, \\ q_{t,-} : \quad & \Pr(Y = 1 \mid F_t = A) = 1 - c, & \Pr(Y = 1 \mid F_t = B) = 0. \end{aligned}$$

Outcomes are conditionally independent across coordinates given F . If $\Pr(F_t = A) = 2/3$, direct substitution in $I(F_t; Y) = H(Y) - H(Y \mid F_t)$ gives

$$I(F_t; Y_{t,+}) = 0.540852 =: I_{\text{hi}}, \quad I(F_t; Y_{t,-}) = 0.459148 =: I_{\text{lo}};$$

the values reverse when the prior is $1/3$.

B.2 PROOF OF THEOREM 1

Place a uniform distribution over the 2^B frames. Until coordinate t is queried, ω_t is independent of every observation from the other coordinates. Thus any frame-blind policy chooses the correct polarity on a fresh coordinate with average probability at most $1/2$, and obtains at most

$$\frac{1}{2}(I_{\text{hi}} + I_{\text{lo}}) = 0.5$$

bits on its first query to that coordinate.

It remains to show that adaptivity and repeated queries do not improve this average bound. The prior entropy of one coordinate is $h(2/3) = 0.918296$ bits. Even after the lower-information first query, its remaining entropy is at most

$$h(2/3) - I_{\text{lo}} = 0.459148 < 0.5.$$

Consequently, even a hypothetical follow-up that revealed all remaining uncertainty is worse than the average value of querying a fresh coordinate. A frame-aware policy also moves to a fresh coordinate

because $h(2/3) - I_{hi} = 0.377444 < I_{hi}$. An exchange argument therefore transforms either optimal policy into one that queries each coordinate exactly once without reducing value.

The frame-aware value is BI_{hi} , whereas the average frame-blind value is at most $B/2$. Hence the average regret over frames is at least

$$B(I_{hi} - 1/2) = \frac{B}{2}(I_{hi} - I_{lo}),$$

so at least one fixed frame attains the claimed lower bound. A policy that visits each coordinate once and independently randomizes its polarity obtains exactly $(I_{hi} + I_{lo})/2$ on every frame, proving tightness.

For Hamming loss, the posterior Bayes error is $1/12$ after the correct-polarity channel and $1/6$ after the wrong-polarity channel. Fair randomization gives $1/8$. An untouched coordinate has error $1/3$, so a fresh query reduces error by $5/24$; all error remaining after a first query is at most $1/6 < 5/24$. The same exchange argument applies, and the minimax excess is

$$B(1/8 - 1/12) = B/24.$$

B.3 PROOF OF THEOREM 2

Let $B \geq 3$ be odd. The five canonical candidates are A, B, C, D, E , and the pool has B good and B bad binary items. On every good item their responses are $(0, 0, 1, 1, 1)$ and the reference is 0. On bad item t their responses are $(0, 0, 1, 0, 0)$, with reference 1 for odd t and 0 for even t . The refinement adds four exact copies of C .

For Select-LLM’s exact-match acquisition, write

$$S(x) = x^2 + (1 - x)^2,$$

where x is posterior mass predicting one. After s clean good labels, A and B have likelihood weight $a = e^{s/\tau}$ and C, D, E have weight one. With $d = 2a + 3$, an unqueried good item has one-mass $3/d$, while a bad item has one-mass $1/d$. Since

$$S(1/d) - S(3/d) = \frac{4(d - 4)}{d^2} > 0,$$

every remaining good item has strictly lower acquisition score. The clean selector exhausts the good block, and A, B remain selected.

In the refined registry, behavior C has five of nine entries. Along the bad path, its aggregate posterior mass is $x = 5/9$ after every even number of bad labels and

$$x = \frac{5e^{1/\tau}}{5e^{1/\tau} + 4}$$

after every odd number, so $1/2 < x < 1$ for every finite $\tau > 0$. A bad item has one-mass x . A good item groups C with D, E and has one-mass $(1 + x)/2$. The bad item is strictly closer to one half because $x - 1/2 < x/2$ whenever $x < 1$, and hence has lower S . Induction shows that the refined selector exhausts the disjoint bad block.

Across the complete $2B$ -item pool, A and B outperform C by

$$\Delta_B = \frac{B - 1}{2B}.$$

After each odd bad label, C has one more observed correct response than every non- C candidate, so the selected behavior must be C . There are $(B + 1)/2$ such steps. Even granting favorable tie-breaking on every even step, cumulative regret is at least

$$\frac{B + 1}{2} \Delta_B = \frac{B^2 - 1}{4B}.$$

The final step is odd, giving terminal regret Δ_B .

B.4 PROOF OF PROPOSITION 1

Consider any allowed refinement and couple the clean and refined runs using the same policy randomization and reference randomness. Initially, fixed root masses and refinement-invariant summaries give identical root-level states. Assume the transcripts agree through step $t - 1$. The selector’s state is then the same by factorization, hence so is its conditional action kernel. Under the coupling it selects the same query and observes the same reference; refinement invariance produces the same updated root summaries and root decision. Induction gives identical complete transcripts and decisions almost surely, hence equal distributions. The argument applies to any acquisition objective—including uncertainty, diversity, or mutual information—only if its implemented state and action kernel satisfy the stated factorization.

B.5 EXECUTABLE VERIFICATION

The standard-library checker verifies the channel information values and Bayes errors; every integer root partition for $B = 1, \dots, 6$; all deterministic polarity vectors against all frames for $B = 1, \dots, 8$; and the amplification instance for $B \in \{3, 5, 11, 51\}$ and $\tau \in \{.1, .5, 1, 3, 10\}$. All assertions pass. At $B = 51$, every tested temperature gives clean cumulative regret zero, refined cumulative regret 12.745098, terminal regret .490196, and 51/51 acquisitions redirected to the disjoint block.

C COMPLETE LLM PROTOCOL AND RESULTS

T2-M target selection used development pools 3500–3999 and was complete over every eligible parent; primary outcomes used paired pools 50000–50499 only after the parent lock. The complete robustness cross is temperature $\{.1, .5, 1, 3, 5\}$, alias count $\{1, 2, 4, 8\}$, response distance $\{.1\%, 1\%, 5\%, 10\%\}$, and the three task-specific budgets in Table 2. This yields 240 MedQA, 180 GSM8K, and 240 OpenBookQA cells. GSM8K’s locked parent is correct on too many items to construct quality-identical 10% aliases, so all 60 such cells are marked ineligible rather than retargeted.

Table 7: Complete T2-M grid; every eligible cell is retained. Positive and negative indicate the sign of mean paired cumulative-regret change.

Task	Eligible	Positive	Negative	BH-significant +/−
MedQA	240	240	0	240/0
GSM8K	180	180	0	180/0
OpenBookQA	240	171	69	167/60

At 5% distance, four aliases, and the primary budget, all five temperatures are positive in all three tasks. At each locked temperature, all three budgets are likewise positive, for 24/24 task–setting cells. OpenBookQA supplies the important non-monotone boundary: at its primary setting, four 5% aliases give +1.460 cumulative regret but eight give −.438. We report all cells because alias count and distance change a directional prior; they are not scalar corruption strengths.

Every final pool is sampled without replacement and shared by its clean/refined pair. The paired bootstrap and sign-flip tests use 10,000 repetitions; Benjamini–Hochberg correction is applied within each prespecified family. Target development/final mean-effect rank correlations are .991/.995/.999 for MedQA/GSM8K/OpenBookQA. The raw artifact stores the 500 paired cumulative and terminal arrays, query paths, selected roots, hashes, and eligibility reason for every cell.

Peer-response-free realism confirmation. After the T2-M targets were frozen, a separately hash-locked audit rebuilt four 5%-distance aliases from only the parent’s outputs, public IDs/options, and references. It used no peer response or response-frequency statistic and generated pools only after construction. Under the historical backend, fresh seeds 91000–91499 gave mean paired $\Delta CReg$ of 3.2657 [3.0341, 3.4997], 3.7851 [3.6247, 3.9492], and .9552 [.8742, 1.0342] on MedQA, GSM8K, and OpenBookQA. Fixed-query paired effects were exactly zero. A sequential-reference variant stopped after 258/1000, 697/1000, and 136/500 inspected labels; it harmed the first two tasks but improved Open-

BookQA by .6805. The preregistration, raw 500-pair arrays, and isolated-constructor replay are stored in the Step 54 artifacts; the validator reports `PASS_STEP54_MATRIX_FREE_REALISM_AUDIT`.

Canonical V2 implementation-stability transfer. An implementation audit found that evidence from a separately tested legacy matrix-aware wrapper could not certify the paper’s matrix-free headline condition. Before computing V2 outcomes for that exact condition, we therefore hash-locked its unchanged aliases, data, pools, targets, budgets, and temperatures, plus canonical entry ordering and an exact 3×3 tie-policy cross. The V2 replay gives 3.2518 [3.0214, 3.4842], 3.7811 [3.6177, 3.9433], and .8459 [.7650, .9260]. Fixed-query effects remain exactly zero, so sequential-excess effects equal the full paired penalties, with independently recomputed positive lower bounds of 3.0231, 3.6209, and .7649. All 16 permutations of both registries produce zero bitwise mismatch over 48,000 scenario–pool paths. Every one of the 27 tie cells changes all paths and has a positive cumulative interval; MedQA/GSM8K terminal means are positive under every tie policy, while OpenBookQA is not promoted as terminal harm. A runner-independent reconstruction replays aliases, acquisition paths, 10,000-repetition inference, BH correction, and adjudication in 2,882 checks, reporting `PASS_STEP80_INDEPENDENT_VALIDATION` and `STRONG_TRANSFER`. Main-text values use this V2 backend; Step 54 values above remain the historical record.

Executable endpoint replay. A subsequent outcome-blind lock converted only the changed coordinates into immutable SHA-256-prompt/output maps. Runtime requests contain question text and ordered public options but no dataset ID, reference, correctness tag, peer response, pool seed, or selector state. Four endpoints per task reproduce all 12 Step 54 response hashes and all 13 selector-run fields exactly on the same 500 fresh pools. Manifests range from 2,272 to 5,088 bytes; appending a probe character to every supported changed prompt triggers zero overrides. This establishes feasibility under output-only research interfaces, not admission to a particular production registry. The independent validator reports `PASS_STEP55_EXECUTABLE_ENDPOINT_REALISM_AUDIT`.

Calibration-to-holdout, holdout-reference-free audit. A preregistered hash split produces labeled calibration halves of size 500/500/250 and blind holdouts of size 500/500/250 with zero ID overlap. Before holdout outcomes, five-fold calibration selected MiniLM/TF-IDF error learners and froze 1%-distance alias rows; the blind builder received only prompts, public options, and parent outputs. All holdout aliases stay within one accuracy point and at most 1% oracle disagreement. Mean $\Delta CReg$ is +.2941 [.1079, .4791], +.1599 [.1059, .2141], and −.1124 [−.1421, −.0835] on MedQA, GSM8K, and OpenBookQA. Fixed-query effects are −.0306/0/0. GSM8K selected-support error precision is zero, so this is small-distance approximate control, not exact quality equivalence. The locked machine decision is `PROMOTE_LABEL_FREE_T1`, where “label-free” is scoped only to holdout construction; the validator reports `PASS_STEP56_CALIBRATION_TO_HOLDOUT_AUDIT`.

D NATURAL EXACT-ALIAS AUDIT

The natural anchor is selected by the unique intersection of exact response-and-oracle pairs across MedQA, GSM8K, and OpenBookQA, before running the base-versus-refined selector comparison. The 30-entry stable-order quotient retains frozen index 9; the original 31-entry roster restores index 10 and maps it to root 9. A primary task is path-material at path-change fraction at least .10 and one mean changed position; it is outcome-material at $|\Delta CReg| \geq .1$, a zero-excluding interval, and three-task BH $q < .05$. Across 21 prespecified temperature/budget cells, all cells are path-material and 14 have family-corrected cumulative-regret differences, split into five harms and nine improvements. Candidate-invariant random acquisition gives exact query, root-path, cumulative-regret, and terminal-regret equality in every primary task. The complete Step 50 arrays are stored at SHA-256 14B77BD3...B0F8EA4.

A subsequent descriptive replay distinguishes changed positions from unique labels. Mean base-only query counts are 4.358, 1.420, and .960; query sets differ in 94.0%, 87.4%, and 66.0% of runs. Mean first divergence occurs at steps 1.486, 3.698, and 4.690, while mean changed root-decision positions are 10.592, 2.902, and 2.012. The 5% replacement threshold is an analyst-chosen heuristic defined after aggregate Jaccards were known, not confirmatory evidence. Raw arrays and exact random controls are stored in `STEP52_OPERATIONAL_SIGNIFICANCE_RESULTS_2026-08-09.json` (SHA-256 BB70CEFD...E372FA).

E CODA OFFICIAL-CODE AUDIT

The official implementation is locked at commit `6c95b7f8`. A prediction-only rule—fixed before labels—selects one parent from the lowest-consensus quartile by the increase in initial family P_{Best} mass after adding four variants. The near variants use $q_j = (1 - \epsilon_j)p + \epsilon_j/C$ for $\epsilon_j \in \{.0005, .001, .0015, .002\}$, preserving every hard prediction and correctness value while changing probability tensors. Exact4 is the multiplicity control. CODA’s EIG configuration and a 30-label budget are fixed; only CoLA activates the public stochastic tie flag, for which all ten frozen seeds are paired.

Table 8: Locked official-code CODA near4 audit. All tasks change their ordered path; exact4 gives the same task-level effects. WNLI ($\dagger$) was not used for target selection.

Pool	Δ cumulative regret	First divergence	Query Jaccard	Δ terminal regret
CoLA	+0.051965	1	0.622	−0.001918
MRPC	+0.053921	5	0.935	0
RTE	+0.841155	2	0.818	0
SST-2	−0.271789	4	0.714	+0.002294
WNLI $\dagger$	−0.014085	1	1.000	0

Near4 changes the ordered path on CoLA, MRPC, RTE, SST-2, and the prospectively unused WNLI pool. Cumulative-regret changes are +.051965, +.053921, +.841155, −.271789, −.014085, respectively. Exact4 gives the same task-level effects. WNLI has query-set Jaccard one and only reorders its acquisitions. Thus the audit supports cross-method path sensitivity, not response-distinct defense evasion or universal harm.

F PREREGISTERED LLM SELECTOR AUDIT

Outcome-blind freeze and intervention. The official LLM Selector repository is locked at commit `15faa47`. Before any judge or weak-judge value was printed, one protocol bound both source-file hashes for all six upstream datasets, included every dataset, and selected the first non-null source-order model as parent. There is no target search. Four aliases inherit the parent’s complete strong-judge outcome row coordinate-wise. Each begins with only the parent’s own ten-weak-judge tensor, then a public SHA-256 rule cycles a frozen 20% of ternary coordinates; it reads no other candidate, strong-judge outcome, pool, acquisition path, or selector state. Realized alias–parent weak-view distances range from 17.25% to 22.25%, while every strong-utility distance is exactly zero. This is a controlled pairwise decision-view refinement over released score matrices, not an executable response wrapper or production-registry submission.

The audit reproduces the official uniform-entry posterior and mean ten-judge expected-entropy acquisition at $(\epsilon_1, \epsilon_2) = (.2, .4)$. For each dataset and seeds 0–999, clean and four-alias registries share a pool of size $\min(400, N)$, budget 30, and deterministic original-root tie rule. Primary terminal regret compares the selected original root with the best original root on that pool. The six-part gate requires coordinate-wise utility identity, exactly zero fixed-query terminal and cumulative effects, path-change rate at least .5, mean terminal harm at least .25 percentage points, positive paired-bootstrap lower bound, and six-task BH-adjusted one-sided sign-flip $q \leq .05$. Two passing datasets were frozen as the promotion threshold; all nulls and improvements remain.

Table 9: Preregistered official LLM Selector audit (1,000 paired pools per dataset). Terminal effects are refined-minus-clean root regret in percentage points. All alias strong-judge rows equal the parent coordinate-wise and every fixed-query effect is zero. A dagger marks the two datasets passing the frozen six-part gate.

Dataset	Δ terminal [95% CI]	BH q	Path $\neq$	Set $\neq$	Root $\neq$	Δ CReg
AlpacaEval [†]	.321 [.194, .453]	.000030	100.0%	99.5%	23.2%	-.245
Arena-Hard	-.370 [-.531, -.213]	1.000	100.0%	99.1%	6.7%	-.379
Bingo [†]	.498 [.374, .623]	.000030	100.0%	98.2%	32.4%	.117
Flickr30k	.183 [-.003, .368]	.05298	100.0%	100.0%	66.7%	-.029
MEDIQA	.000 [.000, .000]	1.000	100.0%	100.0%	.0%	.027
MT-Bench	.000 [.000, .000]	1.000	100.0%	100.0%	.0%	.719

AlpacaEval and Bingo pass, producing the preregistered decision GO_STRONG_CROSS_SELECTOR. The result is directional rather than generic: Arena-Hard significantly improves; Flickr30k’s positive point effect misses both the magnitude and corrected-significance gates; MEDIQA and MT-Bench have terminal effect exactly zero, although their interim cumulative effects are positive. All 6,000 ordered paths change, but query-set and terminal-root changes differ by task. AlpacaEval is especially informative about endpoint choice: terminal regret worsens while cumulative regret improves, so neither metric substitutes for the other.

An independent file that imports no runner verifies all input hashes, reconstructs aliases and every stored pool/statistic, and checks 85,934 invariants. It additionally replays 300 result-independent trajectories from scratch using the literal official entropy equation; all paths, roots, and regrets match. The receipt reports PASS_STEP88_INDEPENDENT_VALIDATION; the preregistration, result, and receipt hashes begin B6303706, 24C3FCDA, and the receipt-bound validator hash 4EEE854C.

G CONFIGURATION-ONLY RAW-INPUT WRAPPER TRANSFER

Outcome separation and endpoint. Step 89 fixed QNLI’s model revision, deterministic calibration/holdout split, complete development grid, one condition, and materiality gate before opening 2,772 sealed outcomes. Four small safetensors files store scalar calibration/configuration values rather than learned neural weights; a CLI accepts only a raw question/sentence pair and emits the parsed hard label with an adapter-specific confidence view. Its 1,000-pool result is negative and remains binding. Step 90 then froze two untouched tasks before any parent prediction: MNLI splits 4,933/4,882 and QQP 20,309/20,121 calibration/holdout items. Per task, all 135 combinations of three roster sizes, three budgets, three temperatures, and five high-confidence trigger fractions were retained; the top 12 pilot conditions were rerun on 300 calibration pools before one configuration was hashed. Runtime wrappers read only raw text and pinned-parent logits, contain no item map, and exactly preserve hard answers. They establish raw-input execution but are configuration-only response variants, not learned adapters or full checkpoints.

Table 10: Sealed executable calibration-adapter audits (1,000 paired pools each). Terminal effects are refined-minus-clean root regret in percentage points. Hard answers and utilities equal the parent coordinate-wise; every fixed-query effect is zero. MNLI/QQP q -values use Holm correction. QNLI is the retained earlier single-task negative result and reports its raw p .

Task	Trigger	Δ terminal [95% CI]	q/p	Path/final-root change	Gate
QNLI	all confidence views	-.068 [-.149, .014]	.948	96.9/41.7%	No
MNLI	top 30% confidence	+.702 [+.586, .823]	2.0×10^{-5}	99.8/32.8%	Pass
QQP	top 20% confidence	+1.270 [+1.201, +1.340]	2.0×10^{-5}	100.0/82.3%	Pass

MNLI and QQP both pass the preregistered eight-part gate: at least .5 terminal point, positive paired-bootstrap lower endpoint, two-task Holm $q \leq .05$, at least 50% path change, coordinate-wise answer/utility equality, exact fixed-query equality, and matching active-minus-fixed harm. QNLI demonstrates the direction is not guaranteed. A validator importing no runner reconstructs all 4,000 active and 2,000 fixed-query trajectories, both bootstrap/sign-flip analyses, Holm correction, and 12,077 invariants; its receipt is PASS_STEP90_INDEPENDENT_VALIDATION. The raw result and

validation receipt are hash-bound. This precursor does not satisfy the learned-checkpoint condition addressed next.

H PREREGISTERED LEARNED-ADAPTER DECOUPLED-VIEW AUDIT

New task, outcome separation, and learned parameters. Before downloading an AG News row, Step 92 fixed dataset revision `eb185aad`, base-encoder revision `12040acc`, the full training recipe, split salt, finite attack grid, 1,000 test seeds, estimands, and nine-part pass gate. SHA-256 assigns the 120,000 public training examples to 83,993 root-model, 24,039 adapter-training, and 11,968 attack-calibration items. The complete 7,600-item test split is input-only outside a separately named sealed-label file. A source audit found no earlier AG News result in the project; SST-2 was rejected before new outcomes because it appeared in prior CODA work. This condition uses exact equality of confidence-coded response views for acquisition but hard-label correctness for posterior evidence and regret. It is therefore a decoupled-view selector audit, not an equation-faithful reproduction of Select-LLM’s single similarity.

Twelve bottleneck classification adapters (100,484 learned parameters each) share a frozen DistilBERT encoder and form the clean registry. Each trains without artificial label corruption on a preregistered 2–100% fraction of the root-model partition. For each of five eligible parents, four contextual temperature adapters are independently optimized with four seeds and bootstraps on the disjoint adapter partition. Every saved attack file contains six nonconstant tensors and 50,817 nonzero learned parameters; the four selected files have distinct SHA-256 values. The learned positive temperature rescales parent logits, so the hard label is invariant on every input by construction. A confidence field above a frozen threshold makes the submitted response views distinct. Runtime reads raw text, frozen hidden state, and parent logits—never an item ID, test label, lookup table, peer output, pool, or selector state.

Stage A retains all 300 parent/ τ /bin/trigger configurations on the attack-calibration partition and reruns the top 12 on 300 pools. It freezes parent 8, $\tau = .25$, eight bins, a 30% trigger target, budget 50, and the full 12-root roster. Calibration terminal harm is .609 points with 100% path change. Only then does Stage B open the test labels once.

Table 11: Preregistered learned-adapter *decoupled-view* audit on 1,000 sealed AG News pools. Confidence-coded responses drive acquisition; hard-label correctness drives posterior evidence and regret. Effects are refined-minus-clean terminal root regret in points; every fixed-query effect is zero.

Task	Params/alias	Pairwise response d	Path/root change	Terminal [95% CI]	AMF [95% CI]	Gate
AG News	50,817	24.1–26.3%	100.0/55.2%	+ .547 [+ .502, + .593]	+ .547 [+ .502, + .593]	Pass

On the complete holdout, realized trigger rates are 30.2–31.5%, pairwise alias-response Hamming distances are 24.1–26.3%, and all 30,400 adapter–item hard answers equal the parent. Ordered paths change in 100% of pools, query sets in 99.3%, and terminal roots in 55.2%. Terminal harm is +.54675 points with paired-bootstrap interval [+ .50225, + .59250] and one-sided sign-flip $p = 9.9999 \times 10^{-6}$; active-minus-fixed has the same mean and a positive interval. Fixed-query root histories and terminal/cumulative effects are byte-identical. The locked decision is `GO_SCORE_CHANGING_LEARNED_ADAPTER_BRIDGE`.

A validator importing no experimental runner replays all 2,000 active and 2,000 fixed-query trajectories, reconstructs the statistics and gate in 4,170 checks, and runs 256 raw-input end-point samples through all 12 clean checkpoints and four learned adapters. Its receipt is `PASS_STEP92_INDEPENDENT_VALIDATION`. An initially detected 5.6×10^{-17} acquisition-tie difference motivated a disclosed post-confirmatory audit: sorted, first-appearance, reverse-code, and 10^{-12} tolerance policies change many individual paths but all retain 100% clean–refined path change, exact fixed-query equality, and terminal harm .54575–.54800 points with lower endpoints at least .50125. This robustness audit does not replace the preregistered primary result.

The result closes a learned-parameter, new-task, no-test-lookup bridge for this explicitly decoupled-view selector. It does not establish learned source-faithful Select-LLM terminal harm, independent

full-model fine-tuning, live registry admission, secret-task compromise, or deployment prevalence. The next audit tests the single-similarity gap directly.

I PREREGISTERED SOURCE-FAITHFUL LEARNED CONFIRMATION

Before downloading a Banking77 row (Casanueva et al., 2020), Step 93 fixed the processed dataset revision 18072d26, DistilBERT revision 12040acc, hash partitions, 12-root training recipe, seven eligible parents, 504-condition Stage-A grid, 1,000 test seeds, and twelve confirmatory gates. Two pre-outcome amendments record that the pinned executable mirror contains 9,993/3,076 rather than the original metadata’s 10,003/3,080 rows and includes an unused `label_text` column. Both Stage-0 attempts stopped at assertions before row iteration or output; no scientific choice changed. SHA-256 then assigned 6,015/1,996/1,982 training rows to root training, error-predictor training, and calibration. Test texts and labels were separately hash-bound.

Twelve 109,901-parameter intent adapters share a frozen encoder. For each of seven eligible parents, four independently bootstrapped 50,817-parameter networks learn to predict parent error from the hidden vector. A selected alias emits either the parent’s integer label or its unique literal `ABSTAIN_a`, which is outside all 77 references. Hence its exact-match evidence is coordinate-wise no greater than the parent’s without consulting a test label. The runner accepts response codes and references but no external feedback/utility matrix: the identical function $s(a, b) = \mathbf{1}[a = b]$ constructs candidate-equality groups and reference feedback. The execution lock binds both call sites’ source hashes, all 40 weight files, the complete calibration ledger, sealed labels, and code before output.

Table 12: Preregistered learned single-similarity confirmations on sealed Banking77 (1,000 pools) and selected fresh 20 Newsgroups (2,000 pools). The same exact-match s drives acquisition and posterior evidence. Loss and terminal columns are percentage points; CReg is the 50-step cumulative root-regret difference. Fixed-query effects are exactly zero.

Task	Params/alias	Alias loss	Path/root	Δ Terminal [95%]	Δ CReg [95%]	Gate
Banking77	50,817	.098-.358	99.8/36.6%	+.032 [-.055, +.122]	+.0409 [+ .0129, +.0689]	No
20 Newsgroups	205,245	.252-.491	100/48.2%	+.120 [-.044, +.290]	+.0822 [+ .0385, +.1266]	No

The selected parent is root 10 with $\tau = .1$, budget 50, and a .25-point calibration-loss cap. On the 3,076-item holdout, aliases lose .098-.358 accuracy points, trigger on .10-.52% of items, and are coordinate-wise non-improving. Across 1,000 paired pools, 99.8% of ordered paths, 99.1% of query sets, and 36.6% of terminal roots change. Cumulative regret rises by +.040945 [+ .012905, +.068910], but terminal regret rises only +.032 [-.055, +.122] percentage points with one-sided $p = .236$. Active-minus-fixed terminal harm is identical; all fixed-query root histories and terminal/cumulative effects are exactly equal. The preregistered .5-point terminal gate therefore fails and the retained decision is `NO_GO_RETAIN_SOURCE_FAITHFUL_NEGATIVE`.

The first machine result printed `INVALID_STEP93_IMPLEMENTATION` because a supplemental dense-versus-sparse acquisition check required absolute error at most 2.0×10^{-16} ; the two algebraically equal reductions differed by 2.220446×10^{-16} , one binary64 ULP at one. Source hashes, bitwise reference feedback, endpoint replay, coordinate constraints, trajectories, and all scientific gates were unaffected. An independent implementation reconstructs all 1,000 clean/refined active and fixed runs, inference, and 128 raw-input endpoints in 82 checks and adjudicates the valid retained negative without promoting any gate or changing an outcome. No replacement task is opened.

J FRESH-TASK LEARNED SINGLE-SIMILARITY REPLICATION

After retaining the Banking77 terminal negative, Step 94 fixed a finite menu—20 Newsgroups, MASSIVE en-US intent classification, and SST-5—before materializing their test splits. SHA-256 partitions each development population into root training, gate training, selector search, and selector verification. Sixteen independently trained bottleneck classifiers per task form a candidate bank. Three prespecified quality-gap templates deterministically select eight-root rosters from search accuracy; only the top three search roots are eligible parents. Four independently bootstrapped

error gates per eligible parent consume frozen CLS vectors and parent logits. An integrated variant checkpoint contains a copy of the parent classifier plus one learned gate and accepts raw text only. Above its frozen threshold it emits a unique out-of-alphabet ABSTAIN_a; otherwise it emits the parent class, guaranteeing coordinate-wise non-improvement without test outcomes.

The complete search comprises 648 conditions per task. The top 24 are rerun on 500 disjoint development-verification pools, after which a locked rule selects the task with largest positive active-minus-fixed terminal effect among eligible tasks. All three are eligible; verification effects are $+ .727 / + .208 / + .134$ terminal points for 20 Newsgroups/MASSIVE/SST-5, with 100% path change and exact fixed-query equality. The rule selects 20 Newsgroups, root 10, the eight-root roster (10, 9, 6, 3, 1, 0, 2, 4), $\tau = .25$, and four thresholds. A lock binds the complete 1,944-cell search ledger, 72 verification rows, 128 learned files, code, the selected test, and 2,000 confirmatory seeds before only the selected test label file is opened. The other two test outcomes remain unanalyzed.

Each selected integrated checkpoint has 205,245 nonzero learned parameters and a distinct SHA-256 digest. Raw-input replay exactly reproduces the parent component and never accepts a label, item ID, lookup, peer output, pool, trajectory, posterior, or selector state. On 7,532 sealed test items, parent accuracy is 60.847%; aliases lose .252–.491 points and trigger on 1.53–2.20%. Every ordered path and query set changes, as do 48.2% of terminal roots. Cumulative regret increases by $+ .082157 [+ .038544, + .126554]$ with one-sided sign-flip $p = .000200$. Terminal regret increases by only $+ .1198 [- .0444, + .2895]$ percentage points with $p = .0813$, below the frozen .5-point gate; active-minus-fixed is identical. Fixed-query root histories and both regret arrays are bitwise equal. The retained decision is NO_GO_RETAIN_FRESH_TASK_NEGATIVE: a second source-faithful learned cumulative result, not a terminal success.

An independent validator reads the locked raw arrays rather than the experimental runner and reconstructs hash bindings, all delta vectors, coordinate non-improvement, quality losses, bootstrap intervals, sign-flip values, path changes, fixed-query equality, every gate, and the retained decision. All checks pass. This audit strengthens replication and raw-endpoint realism for the paper’s primary anytime endpoint, but does not close the stronger independently fine-tuned full-checkpoint or material terminal-harm condition.

K MULTINLI FULL-ADAPTER DEVELOPMENT STOP

After Step 94’s sealed terminal negative, Step 95 prospectively fixed one new task rather than reopening that menu: MultiNLI (Williams et al., 2018), revision da70db2a, with labeled training data partitioned by genre and class and `validation_matched` stored as separate input and sealed-outcome files. Eight public NLI checkpoints and revisions were frozen before label-dependent evaluation. A deterministic rule selected the strongest of three eligible DistilBERT parents and five additional roots. For each of four aliases, a separately seeded binary classifier learns parent error from premise–hypothesis text while freezing embeddings and the first four DistilBERT blocks; the last two blocks, pre-classifier, and classifier comprise 14,767,874 learned parameters. The runtime endpoint accepts only the two raw text fields and emits the parent label or a unique out-of-class abstention token. The same literal exact-match function supplies output grouping and reference feedback.

The complete 48-cell grid crosses three temperatures, four correct-coordinate loss caps, and four trigger caps on 400 search pools; its top eight cells are replayed on 1,500 disjoint verification pools. The frozen winner uses the `typeform` parent, six roots, $\tau = .05$, a 1-point threshold loss cap, and a 5% trigger cap. On verification, aliases lose .724–.895 accuracy points and trigger on 1.01–1.22% of examples. Ordered paths change in 99.33%, query sets in 99.20%, and terminal roots in 21.13%. Cumulative regret rises by $+ .050557 [+ .042609, + .058891]$, while terminal and active-minus-fixed regret rise by only $+ .04987 [+ .02640, + .07307]$ percentage points. Fixed-query effects are exactly zero and root histories are bitwise equal.

The positive terminal interval does not meet the preregistered +.5-point materiality gate; search terminal harm is only $+ .0375$ points, and three search aliases exceed the one-point quality cap (maximum 1.3867 points). The locked decision is therefore NO_GO_DEVELOPMENT_GATE_STOP. No confirmatory execution lock was created, `validation_matched` outcomes were not opened, and `validation_mismatched` was not substituted. An independent validator replays the selected 400 search and 1,500 verification trajectories, reconstructs the complete-grid/top-eight selection,

Table 13: SNLI learned single- s audit on 3,000 sealed-test pools. Effects are refined-minus-clean; terminal units are percentage points. The locked primary decision remains NO_GO. The conservative row is a complete post-confirmatory sensitivity over the two unique thresholds represented in all eight predeveloped top conditions; its q is Holm-adjusted across those two.

Condition	Alias loss pp	Path/final-root	Terminal pp	CReg	Fixed	Status
Locked primary	.875–1.018	99.13/16.23%	+1.631 [+1.360, +1.903]	+.316 [+.285, +.347]	0	No-go
Conservative top-8 sensitivity	.316–.478	99.43/13.23%	+.915 [+.667, +1.173]	+.221 [+.193, +.250]	0	Secondary pass

inference, all gates, decision, hashes, and absence of confirmatory output. This third learned single- s audit reinforces path and anytime sensitivity but not a material terminal or operational bridge.

L SNLI DIRECTIONAL LEARNED TERMINAL AUDIT

Development-only diagnosis and prospective lock. A post-hoc analysis of Step 95’s frozen verification partition, explicitly excluded from confirmatory evidence, found that 21.13% terminal-root change produced only +.04987 points because the final contest was almost entirely between roots with a 1.01-point global gap and transitions occurred in both directions. Conditional on a changed root, the net drop was only about .22 points. An oracle geometry calculation was used only to establish design feasibility: a strong parent plus meaningfully lower challengers could convert directional transitions into material terminal cost. Before any SNLI row or outcome was inspected, Step 96 then froze SNLI (Bowman et al., 2015), revision `cd5c3d`, a four-root registry, budget 10, three temperatures, four construction-loss caps, four trigger caps, deterministic train partitions, disjoint official validation, and one sealed official test. The public DeBERTa parent exceeds the three frozen challengers by 17.05 points on validation. Four independently seeded raw-text error adapters train only on SNLI train labels; each saves 14,767,874 learned parameters. One literal exact-match function supplies both agreement and reference feedback.

Development authorization and integer-boundary repair. On 1,500 validation pools, the selected cell changes 98.73% of paths and moves the terminal root from the parent to a challenger in 14.8% of runs versus 2.93% in reverse. Terminal and active-minus-fixed harm are +2.231 [+1.836, +2.629] points; CReg is +.420 [+.376, +.465], and fixed-query histories/effects are exactly equal. The first gate computation nevertheless emitted NO_GO because exactly $45/4500 = 1\%$ was represented as 0.010000000000000009. Before test access, an amendment fixed only this numerical implementation to compare integer loss counts: threshold/search/validation maxima are 45/4500, 57/6000, and 86/9842, all within the locked one-point cap. Original ledgers are retained, the chosen cell and every effect remain unchanged, and a receipt binds the repair before the one-time test lock.

Locked primary test and complete top-eight sensitivity. The label-free test predictions of all roots/adapters were hashed before the sealed test outcomes opened. On 3,000 locked pools, the selected primary changes 99.13% of paths and 16.23% of terminal roots. Terminal and active-minus-fixed harm equal +1.631 [+1.360, +1.903] points ($p = 9.9999 \times 10^{-6}$), and CReg is +.316 [+.285, +.347]; fixed-query effects are exactly zero. Yet the primary remains NO_GO_RETAIN_STEP96_NEGATIVE: one alias loses 100 correct examples, two above the allowed 98, or 1.018% instead of at most 1%. The other aliases lose 86–93.

After retaining that decision, we evaluated all eight predeveloped top conditions rather than choosing an unreported alternative. Because trigger caps were inactive, these collapse to two unique threshold vectors. Holm correction is applied across both. The conservative .5-point-construction threshold loses 31–47/9,824 examples (.316–.478 points), changes 99.43% of paths, and satisfies every pre-specified scientific gate: terminal and active-minus-fixed +.915 [+.667, +1.173] points with Holm $q = 1.99998 \times 10^{-5}$, CReg +.221 [+.193, +.250], parent-to-challenger directionality, and exact fixed-query equality. Table 13 reports both rows. This secondary audit is explicitly post-confirmatory and does not replace the primary selection, but all models, adapters, thresholds, and label-free test predictions were fixed before test opening.

An independent validator reconstructs the pre-outcome/test binding, all 3,000 pools and trajectories, effect arrays, bootstrap and sign-flip inference, integer quality counts, gates, and literal primary

NO_GO decision from raw arrays. A final prospectively fixed SciTail (Khot et al., 2018) replication used a single conservative threshold condition, but stopped before adapter training because the frozen parent’s preregistered roster-gate advantage was only .4 rather than 2.5 points; its test remains sealed. No replacement task is authorized. The evidence therefore closes neither a live-registry nor full-base-checkpoint claim; it does add material learned raw-input single-*s* terminal evidence under a targeted offline registry, strongest in a fully disclosed secondary analysis.

M IMDB DEVELOPMENT-INFORMED HELD-OUT SAME-SIMILARITY PRIMARY

Retained failures before the successful lock. After SciTail stopped, Step 98 selected ANLI R3 (Nie et al., 2020) from training data alone. Its development core effects were positive, but the clean parent was terminal in only 53.125% of search pools versus a frozen 60% auxiliary gate, so the development outcome remained unopened. Step 99 preserved the same models and thresholds and prospectively removed only that auxiliary condition before opening the already sealed outcome; terminal harm was $-.0667$ points and one alias lost 14 rather than at most 12 correct examples. Its decision remains NO_GO_RETAIN_STEP99_NEGATIVE. These attempts are not pooled with the later result.

Step 100 then fixed the processed IMDB revision, within-label SHA-256 split, and separate label-free input and outcome files before analyzing candidate predictions. Of the official 25,000 test reviews (Maas et al., 2011), 12,000 form a development population and 13,000 form the one-time held-out population. The initial fixed parent and Step 101’s outcome-blind public-checkpoint screen both fail the development roster-gap gate. Step 102 replaces only the development roster through a disclosed public-card screen, trains four independently bootstrapped error adapters, and retains all 81 search cells. Although every cell meets search eligibility, the search-maximizing cell falls from $+1.878$ terminal points on search to $+.012$ on disjoint verification, producing NO_GO_STEP102_LEARNED_DEVELOPMENT_STOP.

Robust development selection before held-out access. Step 103 freezes a robustness rule before evaluating the other cells on verification: among cells satisfying every gate on both development partitions, maximize the smaller terminal effect, then cumulative effect and path change under fixed tie rules. The 81 nominal cells contain 27 distinct threshold–budget–temperature conditions; all are evaluated and six are dual-eligible. The selected cell uses budget 5, $\tau = .05$, and the .75-point construction-loss threshold. It produces $+1.560$ terminal points on 400 search pools and $+.533$ [$+.419, +.648$] on 1,500 verification pools; fixed-query effects are zero in both. An independent development validator reconstructs the complete grid, equivalence classes, selection, raw arrays, inference, and authorization while confirming that the 13,000 outcomes remain unopened.

Pre-outcome binding and one-time primary. The four selected adapters each contain 14,767,874 learned parameters and have distinct SHA-256 digests. They freeze a public RoBERTa encoder’s lower blocks and learn parent error from development review text; at runtime an endpoint accepts raw text only and emits the parent label or its adapter-specific out-of-label abstention token. It receives no item ID, held-out reference, outcome, peer output, pool, trajectory, posterior, or selector state. The same literal exact-match function groups candidate outputs and scores acquired references.

Before the held-out outcome opens, Step 104 recomputes and binds all four root rows and all four adapter rows on the 13,000 label-free texts. Their trigger counts are 116/126/108/136. The lock additionally binds the development authority, model revisions, adapter weights and thresholds, outcome/input hashes, code, budget, temperature, and seeds 1,003,000–1,005,999. Only then is the sealed outcome opened once for the fixed analysis. The outcome hash remains identical to the Stage-0 seal.

Table 14: Development provenance and disjoint IMDB held-out primary. Intervals are paired 95% bootstrap intervals; development search is descriptive. Path/root gives the percentage of changed ordered paths/final roots.

Partition	Rows	Runs	Terminal Δ (pp)	Δ CReg	Path/root (%)	Fixed	Status
Development search	1,000	400	+1.560	+ .06718	99.75/66.00	zero	selection input
Development verification	2,000	1,500	+ .533 [+ .419, + .648]	+ .02141 [+ .01715, + .02560]	99.80/53.93	zero	lock gate pass
Sealed held-out primary	13,000	3,000	+ .702 [+ .615, + .788]	+ .02795 [+ .02467, + .03129]	99.93/60.67	zero	GO

The parent accuracy is 94.131%, versus 91.008/91.100/89.485% for the three challengers. Alias loss counts are 60/68/57/68 of 13,000, below the allowed 130, and every changed coordinate is parent-correct to abstention; no alias improves a coordinate. Clean/refined parent-selection rates are 63.63/40.33%; parent-to-challenger transitions exceed the reverse 35.43/12.13%. Terminal and active-minus-fixed regret both rise by +.7024 points, with intervals [+ .6155, + .7882] and [+ .6124, + .7891] and one-sided sign-flip $p = 9.9999 \times 10^{-6}$. Cumulative regret rises by +.027947 [+ .024669, + .031293]. Fixed-query root histories and terminal/cumulative arrays are bitwise equal. All sixteen frozen gates pass, so the literal decision is GO_STEP104_IMDB_HELDOUT_LEARNED_SAME_S_TERMINAL_PRIMARY.

The result is a development-informed held-out confirmation: task, roster, and condition were developed on the disjoint 12,000 rows, rather than preregistered before any IMDB data. It closes the narrower no-heldout-reference, learned raw-input, same-view, terminal-mediation condition left by the Banking77 negative. It does not establish an independently trained full base checkpoint, hidden-task transfer, live-registry admission, or production prevalence.

N YELP PROSPECTIVE CROSS-BENCHMARK ONE-SHOT AUDIT

Before-target protocol and admission rule. Before any Yelp Polarity row or model weight was accessed, Step 105 fixed dataset revision bbf1c97a, the complete official 38,000-row test, and a public admission rule (Zhang et al., 2015). The four largest-download, native binary Transformers checkpoints tagged for Yelp Polarity form the registry; the download-rank-one checkpoint is the parent. Reported model-card accuracy is recorded but never used for admission or parent selection. SHA-256 assigns 6,000/1,000/2,000 official-training rows to adapter training, threshold calibration, and threshold safety. There is no selector-search or selector-verification partition, no target-task grid, and no replacement task or second outcome attempt.

Inherited raw-input pipeline and locks. The four independently seeded error adapters inherit Step 104’s architecture and training recipe: each freezes the lower RoBERTa blocks and learns 14,767,874 parameters from raw text and parent error. A stable threshold rule may use only the calibration and safety partitions. At runtime an alias emits the parent hard label or its own out-of-label abstention token and receives no test reference, item ID, lookup table, peer output, pool, posterior, trajectory, selector state, split, or dataset ID. One literal exact-match function supplies both acquisition and reference evidence. Before the official test outcomes opened, a pre-data lock bound the protocol, revisions, code, budget 5, $\tau = .05$, and seeds 1,053,000–1,055,999; a second lock bound all four roots, adapter weights, thresholds, and label-free root/error/alias predictions. The outcome then opened once.

Table 15: Prospective Yelp one-shot result (3,000 paired pools). Effects are refined minus clean. The literal decision remains NO_GO because the terminal means miss the frozen +.5-point gate.

Endpoint	Effect	Paired 95% interval	Frozen gate
Terminal regret (pp)	+ .3527	[+ .3090, + .3977]	$\geq +.5$: no
Active-fixed terminal (pp)	+ .3527	[+ .3076, + .3956]	$\geq +.5$: no
Cumulative regret	+ .017925	[+ .016163, + .019706]	lower > 0: yes
Ordered-path change	99.90%	—	$\geq 50\%$: yes
Fixed-query effect	.0000	exact	= 0: yes

The parent accuracy is 97.797%, 2.103 points above the best challenger. Alias loss counts are 178/209/143/213 of 38,000 (.376–.561 points), below the allowed 380, and no alias improves a coordinate. Across 3,000 paired pools, 99.90% of ordered paths, 99.87% of query sets, and 41.70% of final roots change. Parent-to-challenger transitions exceed reverse transitions 26.53/9.37%. Terminal and active-minus-fixed regret both rise by +.3527 points, with intervals [+ .3090, +.3977] and [+ .3076, +.3956] and one-sided sign-flip $p = 9.9999 \times 10^{-6}$. Cumulative regret rises by +.017925 [+ .016163, +.019706]. Fixed-query root histories and both regret arrays are bitwise equal.

All integrity, quality, path, direction, fixed-query, and inference gates pass. Only the predeclared requirements that terminal and active-minus-fixed means each reach +.5 point fail. The binding decision is therefore `NO_GO_RETAIN_STEP105_PROSPECTIVE_NEGATIVE`; the threshold is not relaxed and no replacement outcome is authorized. A runner-independent reconstruction exactly reproduces every array, interval, gate, and the literal `NO_GO`; an independently written raw-input validator re-infers every root and alias from the 38,000 raw reviews. The result is a statistically directional, no-target-selector-tuning transfer and a preregistered materiality near-miss, not a second primary success. Because both IMDB and Yelp are binary sentiment, it establishes cross-benchmark and cross-registry transfer within one task family, not broad task-family or live-registry generality.

O PROSPECTIVE UNTARGETED-REFINEMENT STRESS TEST

After the targeted experiments were complete, we prospectively locked 14 previously unused tasks: seven official MODEL SELECTOR tasks and seven HELM MATH subjects. The audit crossed exact, near, and independent-diverse additions, producing 420 nonclean task-condition units with 32 paired runs each. All response-only scores and a 50% high-risk set were sealed before outcome access. Only 35/420 effects crossed the preregistered materiality threshold: one beneficial MODEL SELECTOR effect and 34 Select-LLM effects, of which eight were harmful and 26 beneficial. Exact quotienting neutralized all exact additions, and authenticated lineage neutralized every registered exact and near addition by construction.

The attempted ex-ante registry-fragility audit did not transfer. Its magnitude Spearman correlation was .185 with task-block permutation $p = .416$; AUROC was .719, below a static mass-displacement baseline at .788; and it failed the magnitude, materiality, adaptive-value, cross-selector, and fallback gates. Only the non-vacuity gate passed. The locked decision is therefore `STOP_FRAGILITY_PIVOT`. These results rule out a generic “more entries are harmful” interpretation and any claim that our response-only score predicts attacks. They do not test the separately locked targeted T2-A constructor, whose support is deliberately chosen rather than random. The full protocol, 420 effects, six gates, and 98-check independent validation are stored in the Step 61 artifacts.

P DEFENSE AND MUTATION AUDIT

Table 16: Matched 10% behavioral-cover audit. Clean Δ compares covered clean with uncovered clean; positive is worse. “Extra rm.” counts removals beyond exact quotienting.

Task	Undefended Δ CReg [95% CI]	Cover residual	Clean Δ CReg	Extra rm.	Max span
MedQA	+4.853 [4.603, 5.102]	0	+0.569	1	0.8 pp
OpenBookQA	+1.006 [0.942, 1.069]	0	+1.030	6	5.2 pp
GSM8K	Prospectively ineligible; frozen parent not replaced				

Table 17: Representative post-hoc fixed settings. R is absolute residual CReg divided by the undefended effect over five attacks; Clean G/M/O is Δ CReg (positive is worse). All 21 settings appear in the appendix.

Rule, α	Mean R	Max R	$R < 1$	Clean G	Clean M	Clean O
All three, .050	.824	1.061	3/5	−0.180	+0.351	+0.238
MCCA, .100	.765	1.121	3/5	−0.180	+0.275	+1.018
MCCP, .100	.721	1.125	4/5	−0.180	+0.275	+1.112
class-unif., .132	.656	1.185	4/5	−0.180	+0.386	+1.110
MCCA, .132	.584	1.158	4/5	−0.180	+0.386	+1.273

The exact quotient groups identical full response rows and exactly neutralizes the natural pair and T0 copies. A radius- ρ matched cover joins candidates within normalized response Hamming distance ρ and retains the smallest-index representative of each connected component. At 5%, it neutralizes the corresponding attack with incremental clean CReg 0/0/ - .169 across MedQA/GSM8K/OpenBookQA. At 10%, it also neutralizes the attack, but clean CReg increases by .569 on MedQA and 1.030 on OpenBookQA; the latter merges six natural models spanning 5.2 accuracy points and costs 3.975 terminal points. Authenticated lineage returns exact equality by retaining fixed root mass, but is explicitly an oracle-information comparator.

The soft-weight audit crosses class-uniform, MCCA, and MCCP with $\alpha \in \{.01, .05, .10, .132, .15, .184, .213\}$. Its 63 clean and 105 eligible attack cells are stored without outcome filtering in STEP46_CLONE_ROBUST_SOFT_WEIGHTING_RESULTS_2026-08-08.json (SHA-256 15B693F0...B7B136). The original joint regret/path gate remains 0/105 and all residual path-change fractions equal one. The outcome-first reanalysis instead finds 74 mitigated and 31 worsened attack cells; 24 reach at most 50% residual, 11 at most 25%, and four at most 10%. No fixed setting is non-worsening on all five task-distance attacks. All 21 settings improve clean GSM8K CReg but worsen clean MedQA and OpenBookQA CReg; terminal intervals contain zero in 45/63 cells, with ten significant harms and eight improvements. Table 18 reports every fixed setting. Independent validators reconstruct graph breakpoints, priors, paths, effects, and mutation failures rather than trusting recorded decisions.

Table 18: Complete fixed-configuration soft-weight audit. Mean and maximum R summarize absolute residual CReg ratios over five eligible attack cells; “mit.” counts $R < 1$. Clean columns report Δ CReg on GSM8K/MedQA/OpenBookQA. Final is the largest absolute clean terminal shift across the three tasks, in percentage points.

Rule	α	Mean R	Max R	Mit.	Clean G	Clean M	Clean O	Final pp
class-unif.	.010	.814	1.061	3/5	-.180	+.351	+.338	.023
class-unif.	.050	.824	1.061	3/5	-.180	+.351	+.238	.023
class-unif.	.100	.830	1.153	3/5	-.180	+.275	+.973	.020
class-unif.	.132	.656	1.185	4/5	-.180	+.386	+1.110	.203
class-unif.	.150	.644	1.400	4/5	-.314	+.555	+1.108	.181
class-unif.	.184	.744	1.307	4/5	-.211	+.609	+1.455	.037
class-unif.	.213	.730	1.550	4/5	-.160	+.638	+1.290	.037
MCCA	.010	.814	1.061	3/5	-.180	+.351	+.338	.023
MCCA	.050	.824	1.061	3/5	-.180	+.351	+.238	.023
MCCA	.100	.765	1.121	3/5	-.180	+.275	+1.018	.022
MCCA	.132	.584	1.158	4/5	-.180	+.386	+1.273	.024
MCCA	.150	.637	1.295	4/5	-.314	+.555	+1.172	.037
MCCA	.184	.597	1.259	4/5	-.211	+.609	+2.010	.037
MCCA	.213	.714	1.497	4/5	-.160	+.638	+2.483	.037
MCCP	.010	.814	1.061	3/5	-.180	+.351	+.338	.023
MCCP	.050	.824	1.061	3/5	-.180	+.351	+.238	.023
MCCP	.100	.721	1.125	4/5	-.180	+.275	+1.112	.175
MCCP	.132	.734	1.162	4/5	-.180	+.386	+1.897	.181
MCCP	.150	.785	1.390	4/5	-.314	+.555	+1.823	.169
MCCP	.184	.874	1.304	3/5	-.211	+.609	+2.933	.037
MCCP	.213	1.064	1.584	3/5	-.160	+.638	+3.724	.037

Q ARTIFACT MAP

The artifact root maps every main-text number to an immutable result and an independent validator. The primary families are Step 27 (formal checker), Steps 30–31 (complete T2-M grid and covers), Step 35 (official CODA), Steps 46–49 (soft weights), Steps 50–52 (natural pair), Steps 54–55 (peer-free executable T2-A), Step 56 (within-task T1), Step 61 (prospective untargeted stress), Step 80 (canonical V2 transfer), Step 87 (prospective fresh-task T1), Step 88 (official LLM Selector terminal audit), Steps 89–90 (sealed configuration-only wrappers), Step 92 (learned decoupled-view AG News), Steps 93–99 (learned same- s negatives and sensitivity), Steps 100–104 (complete IMDB development and held-out primary), and Step 105 (prospective Yelp one-shot near-miss). Protocols, ledgers, execution locks, raw arrays, literal status, and validation outputs are preserved. The current manifest binds the revised paper, both final studies’ learned adapters, label-free pre-outcome predictions,

1458 sealed outcomes, primary arrays, and independent reconstructions. Offline validation checks every
1459 bundled byte and anonymity. Public source packages, base weights, and third-party repositories are
1460 not embedded; this is validation closure, not regeneration of exploratory history.
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